

Winning Space Race with Data Science

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15 October 2026

Project Repository:
https://github.com/Mogomotsi99/IBM-Data-Science-Projects/blob/main/SpaceX%20Capstone/README.md

Outline

- Executive Summary
- Introduction
- Methodology
- Insights
- Results
- Conclusion

Executive Summary

Business Problem:

A standard Falcon 9 launch carries an operational manufacturing cost of **$62 million**. Perfecting first-stage booster recovery lowers the net operational expense, creating a competitive economic edge.

Methodological Framework:

Compiled a dataset of 90 flights using REST APIs and Web Scraping. Performed EDA via SQL, Seaborn, and Folium. Designed a production Plotly Dash interactive tool. Tuned four core machine learning classification models.

Core Insights:

Recovery success is highly dependent on booster version and target orbit paths. Logistic Regression, SVM, and KNN algorithms achieved a deployment test accuracy of **83.33%**

Introduction

Project Context:

SpaceX advertises Falcon 9 rocket launches with a cost of **$62 million,** other providers cost upwards of **$165 million**. Much of the savings is because SpaceX can reuse the first stage, while other space launch providers traditionally discard multi-million dollar rocket cores after a single operational ascent.

Project Mandate:

Build a robust, scalable predictive data pipeline to classify launch recovery outcomes (Class 1 = Landing Success, Class 0 = Landing Failure). If we can determine if the first stage will land, we can determine the cost of a launch.

Operational Value:

- Mitigate financial risks for commercial satellite launch insurance bidding.
- Optimise logistics and scheduling for recovery drone ships and landing pad teams.
- Uncover underlying correlations between booster hardware blocks and structural survival.

Section 1

Methodology



Methodology

Executive Summary

- Data collection methodology:
  - Query the official SpaceX REST API & automated web scraping
- Perform data wrangling
  - Execute critical clean-up pipelines
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - Build, tune and evaluate classification models

Data Collection

- **Automated API Pull:**
  - Queried the official SpaceX REST API endpoint (launches/past).
  - Processed nested JSON entities containing booster serials, core logs, and launch pads.
  - Flattened structural data keys into clean columns using Python's pandas library.

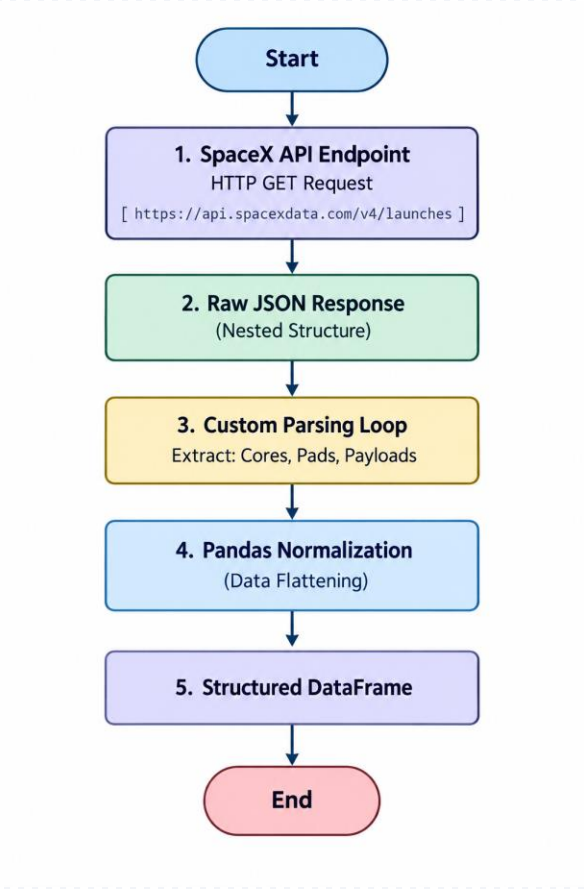
- **Web Scraping Enrichment:**
  - Target Source: Wikipedia's historical Falcon 9 launch table directory.
  - Toolkit: Employed BeautifulSoup to parse HTML tags and isolate missing telemetry variables.

- **Volume:** Consolidated a uniform row count of 90 unique historical flights.

Data Collection – SpaceX API

- **API Endpoint Target:** Consumed the official SpaceX REST API v4 past launches endpoint to extract 90 historical flight records.
- **Payload Request Filtering:** Configured programmatic request parameters to isolate core booster profiles, stripping out irrelevant telemetry vectors.
- **Nested JSON Normalization:** Dissected highly complex, multidimensional JSON payloads containing embedded dictionaries for cores, payloads, and launch pads.
- **Algorithmic Data Flattening:** Applied custom Python mapping functions via Pandas to extract specific keys into a structured tabular matrix.

*Project unavailable due to terminated API server



Data Collection – Web Scraping

- **Target Data Source:** Wikipedia Falcon 9 launch directory web page to capture payload characteristics missing from early automated operational databases.
- **HTML Extraction:** HTTP network requests utilising the Python requests library to download the raw HTML source code structure.
- **Parsing Framework:** BeautifulSoup parsing tree to traverse the web page document object model (DOM), targeting distinct table class layers.
- **Data Extraction & Cleaning:** Iterative processing loop to strip HTML tags from row nodes, extracting explicit flight dimensions including Flight Number, Date, Booster Version, Payload Mass, Orbit, and Launch Outcome

https://github.com/Mogomotsi99/IBM-Data-Science-Projects/blob/main/SpaceX%20Capstone/SpaceX%20Data%20Collection%20with%20Webscraping.ipynb



Data Wrangling

- **Data Sorting:** Filtered rows exclusively down to **Falcon 9** flight vectors.
- **Mean Imputation:** Resolved missing data by replacing null rows within the PayloadMass column with the mean value.
- **Target Variable Derivation:** Mapped landing outcomes into a single binary target column named Class.
- **Outcome Class Mapping Matrix:**
  - Assigned Class = 1 (Landing Success) for controlled outcomes like True ASDS (drone ship) and True RTLS (ground pad).
  - Assigned Class = 0 (Landing Failure) for uncontrolled outcomes, crash events, or lost cores (False ASDS, None ASDS, None None).

https://github.com/Mogomotsi99/IBM-Data-Science-Projects/blob/main/SpaceX%20Capstone/SpaceX%20Data%20Wrangling.ipynb



EDA with Data Visualisation

Visual Data Profiling: Used Seaborn and Matplotlib to build scatter plots and categorical distribution charts, linking continuous numbers against core classification categories.

- Plotted FlightNumber (indicating the continuous launch attempts) againts PayloadMass to visualise how Payload variables affect launch outcome.
- Plotted FlightNumber against LaunchSite to visualise each site's launch records.
- Plotted PayloadMass againts LaunchSite to visualise each site's launches by their payload mass.
- Bar chart visualising the relationship between the success rate of each orbit type.
- Visualised the relationship between FlightNumber and Orbit type.
- Visualised the relationship between PayloadMass and Orbit type.
- Visualised the yearly launch success trend.

https://github.com/Mogomotsi99/IBM-Data-Science-Projects/blob/main/SpaceX%20Capstone/SpaceX%20Exploratory%20Data%20Analysis%20with%20Visualisation.ipynb

EDA with SQL

Relational Database Mapping: Loaded the dataframe into an optimized SQLite database engine to execute high-precision data queries.

- Displayed the names of the unique launch sites.
- Displayed 5 records where launch sites begin with the string 'CCA'.
- Displayed the total payload mass carried by boosters launched by NASA (CRS).
- Displayed the average payload mass carried by booster version F9 v1.1.
- Extracted the date when the first successful landing outcome in ground pad was achieved.
- Extracted the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000.
- Displayed the total number of successful and failure mission outcomes.
- Extracted a list of booster versions that have carried the maximum payload mass.
- Extracted a list of failed drone ship landing outcomes with the corresponding booster version and launch site for months in the year 2015.
- Ranked the count of landing outcomes between 2010-06-04 and 2017-03-20, in descending order.

https://github.com/Mogomotsi99/IBM-Data-Science-Projects/blob/main/SpaceX%20Capstone/SpaceX%20Exploratory%20Data%20Analysis%20using%20SQL.ipynb

Build an Interactive Map with Folium

Geospatial Mapping: Implemented Folium to generate interactive geographical maps with custom marker groupings and vector overlays.

- Added launch site locations on the map using latitude and longitude coordinates.
- Created a Map object with NASA Johnson Space Center at Houston, Texas as the initial centre location.
- Created a Circle object to highlight each launch site and a Marker object to add the launch site name as a popup label.
- Created a MarkerCluster object to group markers containing the same coordinates and applied a Marker to it such that a green marker represents a successful landing and a red marker a failed landing.
- Added a MousePosition to easily find the coordinates of any point of interest on the map.
- Explored the proximity of the KSC LC-39A launch site to railways, highways, coastlines and cities. Added a Marker on each point of interest, displayed the distances using the icon property and drew a line to show the distance using a PolyLine object.

https://github.com/Mogomotsi99/IBM-Data-Science-Projects/blob/main/SpaceX%20Capstone/SpaceX%20Interactive%20Visual%20Analytics%20with%20Folium.ipynb

Build a Dashboard with Plotly Dash

Interactive Dashboards: Built a functional script utilising the Plotly Dash library to render cross-filtered layouts for live data exploration.

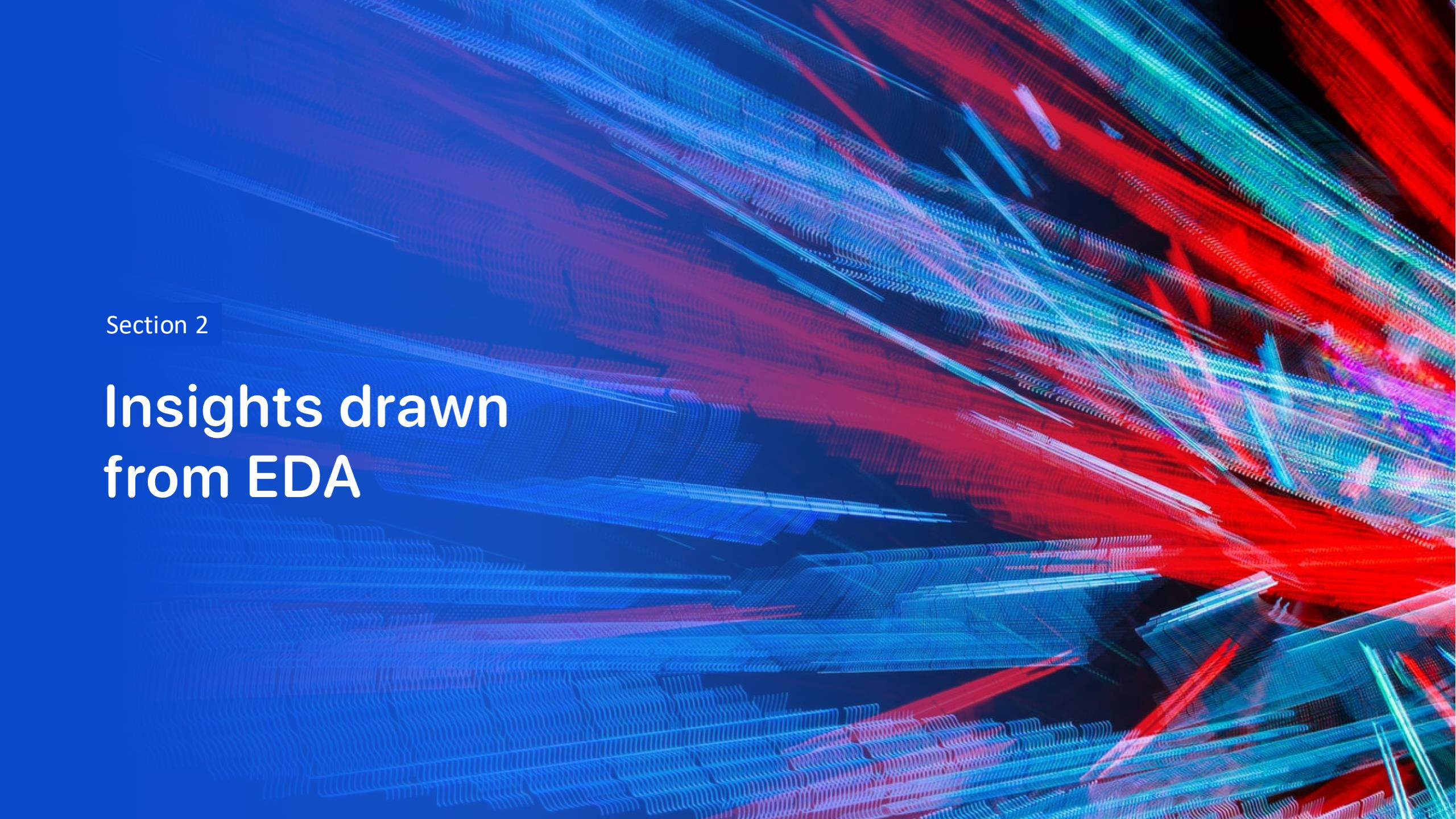
- Dropdown list to enable launch site selection, with all sites as the default select value.
- Pie chart to show the total success launches count for all sites.
- Callback function with the site dropdown as the input and pie chart as the output.
- If a specific launch site is selected, the pie chart will display the success vs failed counts for the site.
- Added a slider to select payload range from 0 to 10 000kg.
- Scatter chart to show the correlation between payload and launch success.
- Callback function with site dropdown and payload slider as the inputs, scatter chart as the output.
- If a specific launch site and payload range is selected, the scatter chart will display the correlation between the selected site's launch success and payload range.

https://github.com/Mogomotsi99/IBM-Data-Science-Projects/blob/main/SpaceX%20Capstone/SpaceX%20Interactive%20Dashboard%20with%20Plotly%20Dash.py

Predictive Analysis (Classification)

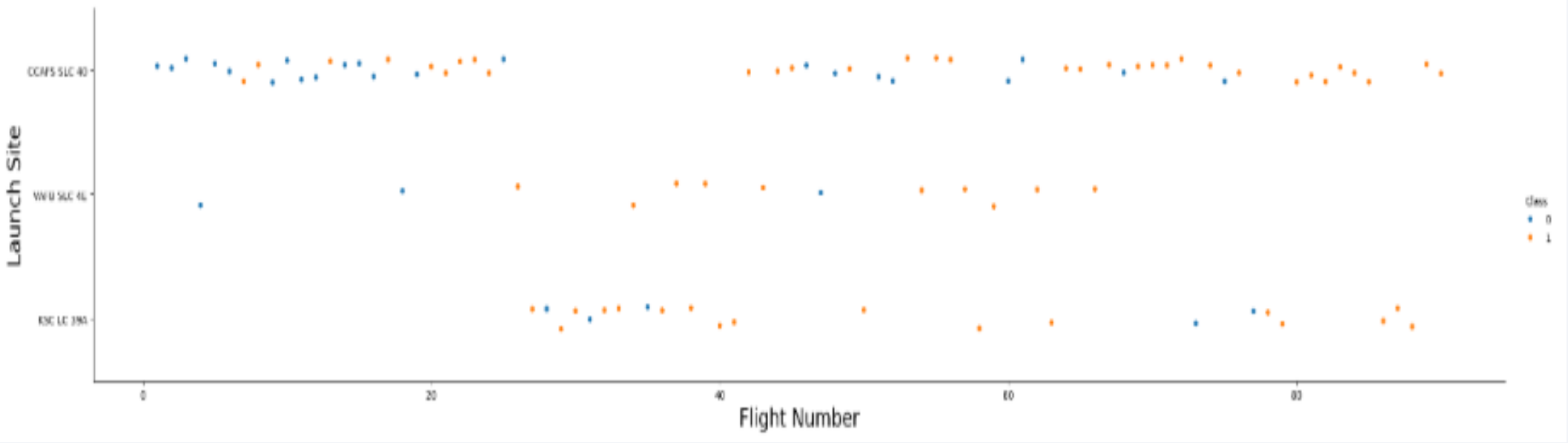
- **One-Hot Encoding**: Converted text fields (Orbit, Launch Site, Landing Pad) into sparse binary vectors via dummy encoding, yielding 83 unique features.
- **Feature Scaling**: Applied StandardScaler transformations to ensure large values (Payload Mass) do not mathematically drown out small binary numbers during distance calculations.
- **Model Selection**: Deployed four supervised algorithms: Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbors (KNN).
- **Hyperparameter Optimisation**: Used GridSearchCV backed by a robust 10-fold cross-validation loop to identify ideal model settings.

https://github.com/Mogomotsi99/IBM-Data-Science-Projects/blob/main/SpaceX%20Capstone/SpaceX%20Machine%20Learning%20Prediction.ipynb


Section 2
Insights drawn from EDA

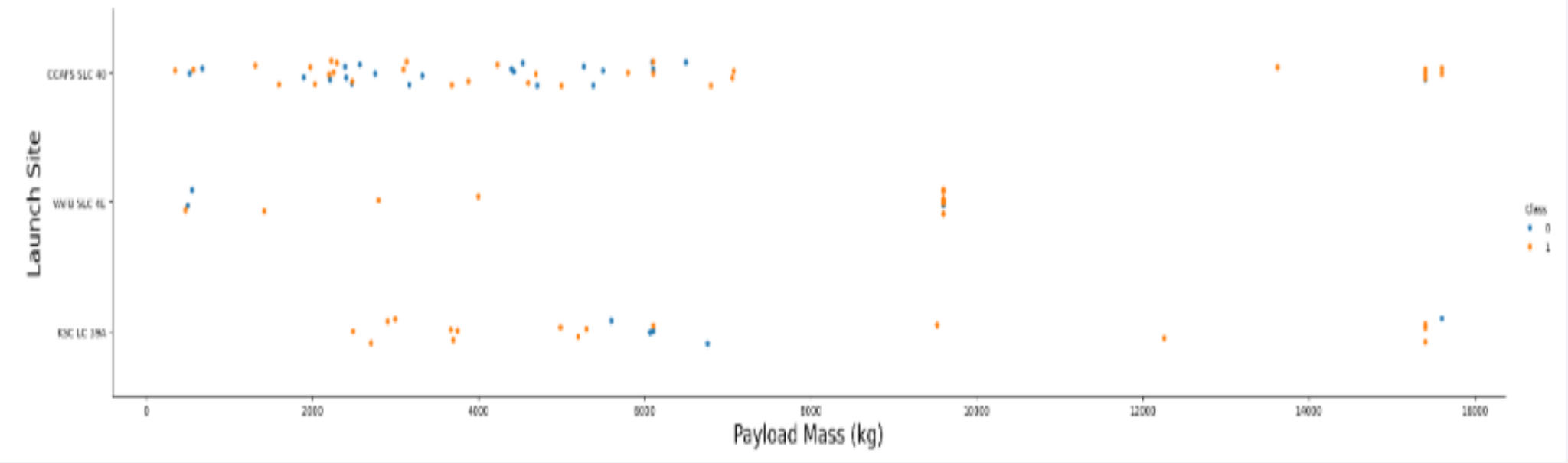
Flight Number vs. Launch Site

- We can observe that CCAFS SLC-40 was used throughout a substantial portion of the recorded launch sequence compared to the other launch sites.



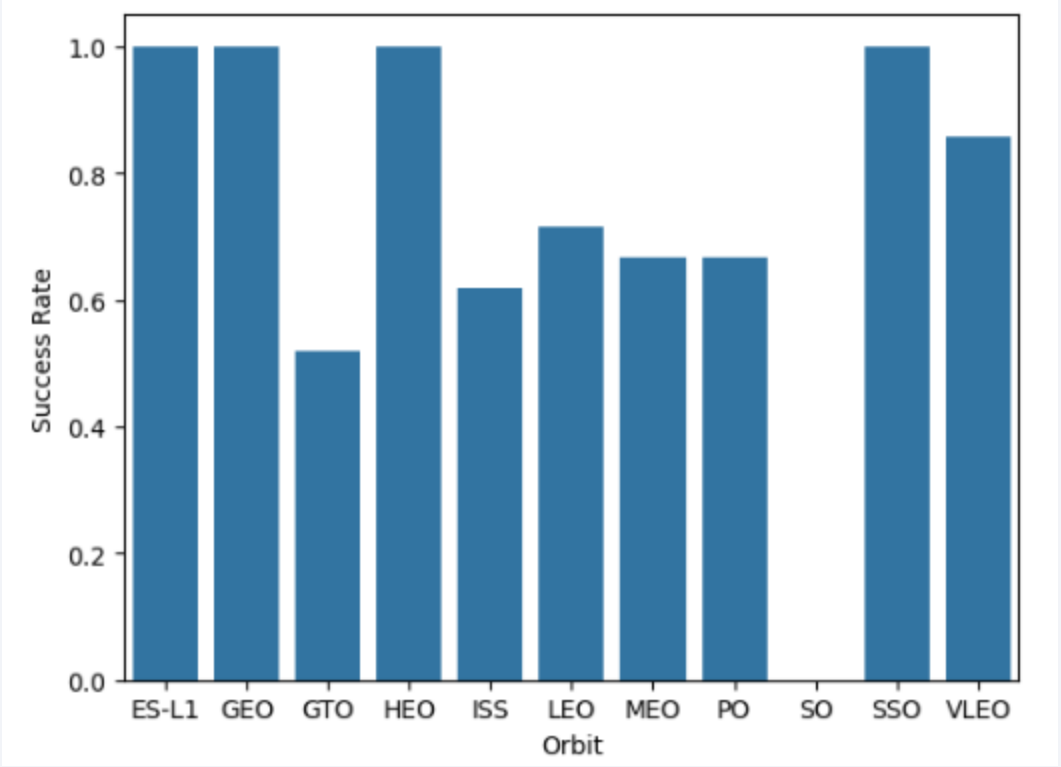
Payload Mass vs. Launch Site

- We can observe that for the VAFB SLC-4E launchsite there are no rockets launched for heavy payload mass greater than 10 000kg.



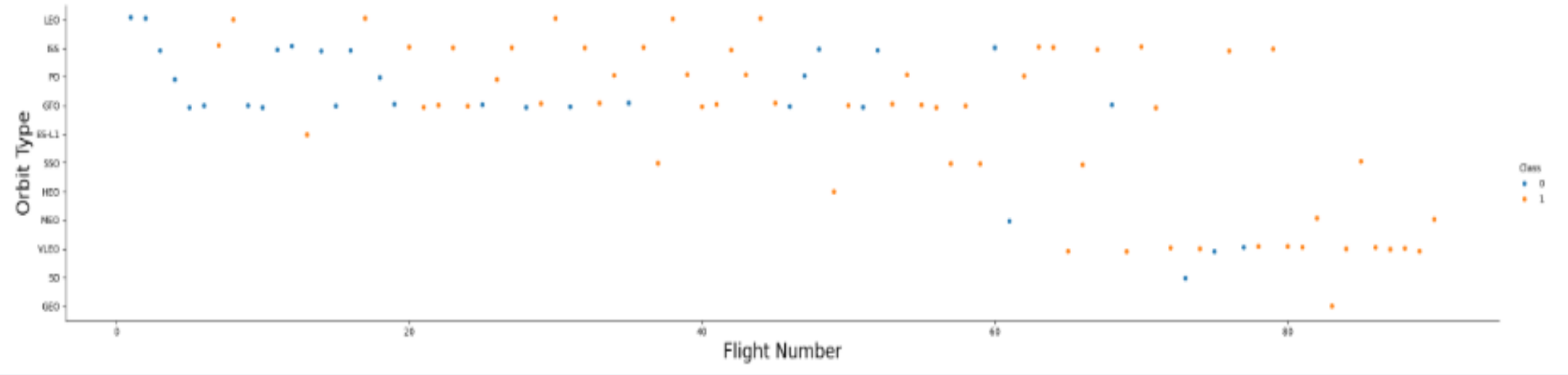
Success Rate vs. Orbit Type

- We can observe that ES-L1, GEO, HEO and SSO maintain the highest success rates.



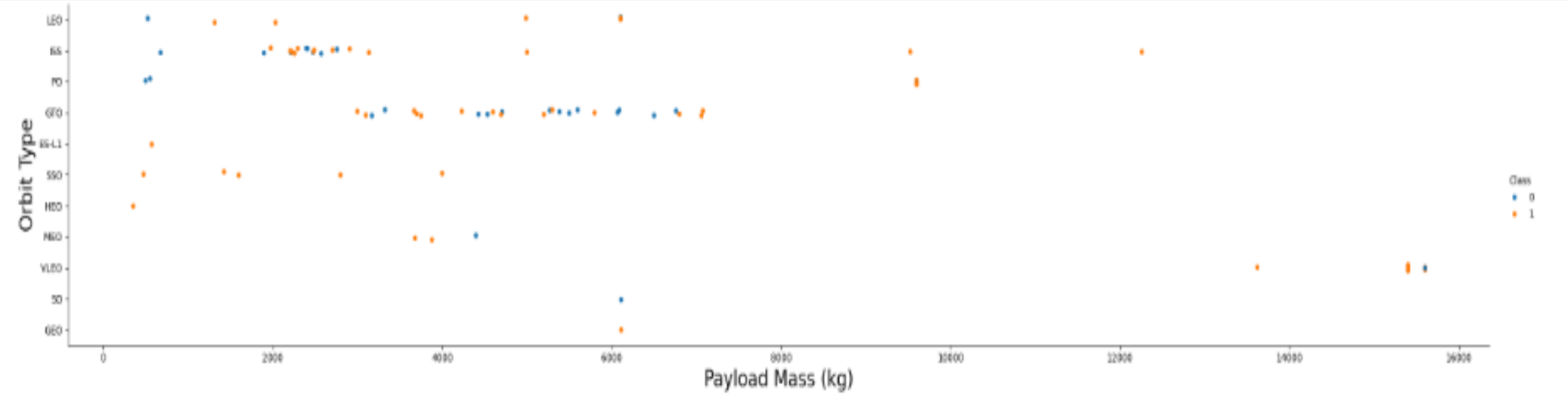
Flight Number vs. Orbit Type

- We can observe that in the LEO orbit, success seems to be related to the number of flights. Conversely, in the GTO orbit, there appears to be no relationship between flight number and success.



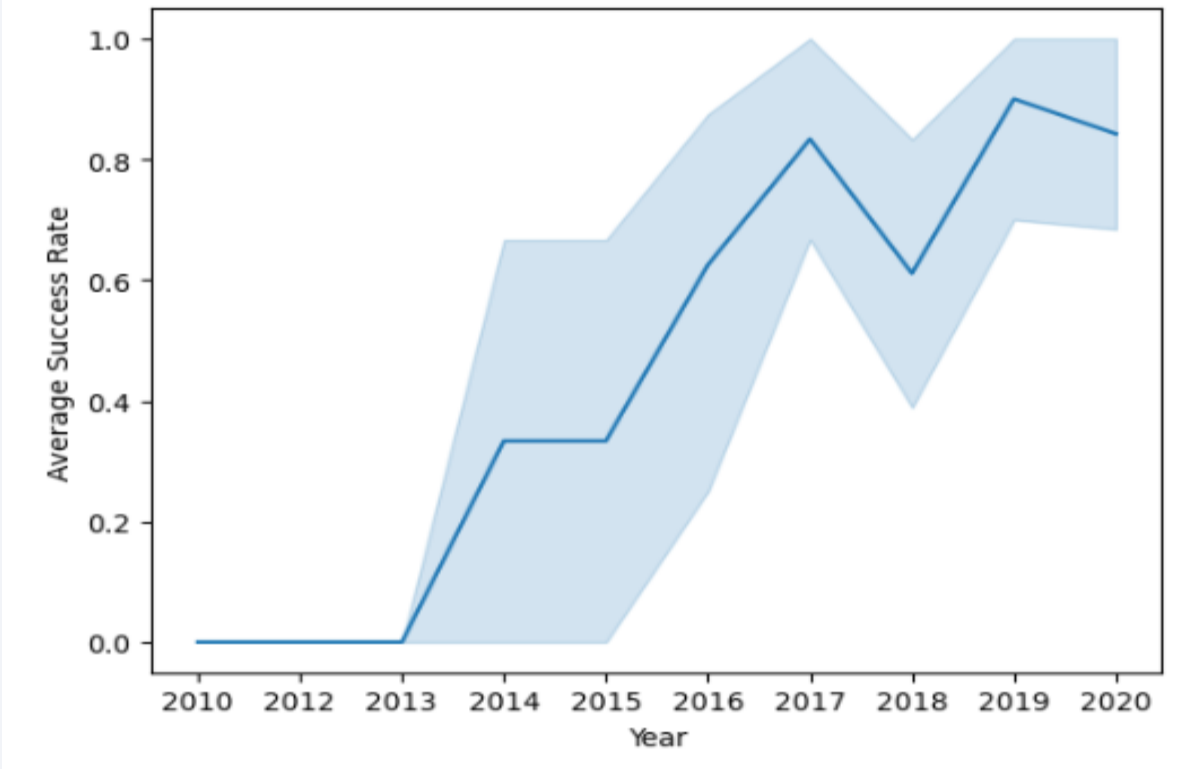
Payload Mass vs. Orbit Type

- We can observe that with heavy payloads the successful landing or positive landing rate are more for Polar, LEO and ISS.
- However, for GTO, it's difficult to distinguish between successful and failed landings as both outcomes are present.



Launch Success Yearly Trend

- We can observe that the annual success rate has been increasing since 2013 until 2020.



All Launch Site Names

- Below are the names of the launch sites contained in our dataset.

```
%sql SELECT DISTINCT "Launch_Site" from SPACEXTABLE;
```

```
 * sqlite:///my_data1.db
Done.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- We can observe that earlier CCAFS LC-40 landing outcomes either failed or were not attempted. CCAFS LC-40 was SpaceX's primary launchpad during the early testing phases of the Falcon 9, therefore it bore the brunt of their experimental trial-and-error period.

```
%sql SELECT * from SPACEXTABLE WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5;
```

```
 * sqlite:///my_data1.db
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- Total payload mass carried by boosters launched by NASA (CRS).
- This reflects a high-frequency cadence of low-to-medium mass deliveries rather than bulk single-launch payloads.

```
%sql SELECT SUM(PAYLOAD_MASS__KG_) AS TOTAL_PAYLOAD_MASS__KG_ from SPACEXTABLE WHERE Customer = 'NASA (CRS)';
```

```
 * sqlite:///my_data1.db
Done.
```

TOTAL_PAYLOAD_MASS__KG_
45596

Average Payload Mass by F9 v1.1

- Average payload mass carried by booster version F9 v1.1.
- The average reflects a heavy reliance on early NASA ISS resupply profiles rather than maximum commercial lift capabilities.

```
%sql SELECT AVG(PAYLOAD_MASS__KG_) AS AVG_PAYLOAD_MASS__KG_ from SPACEXTABLE WHERE "Booster_Version" LIKE 'F9 v1.1%';
```

```
 * sqlite:///my_data1.db
Done.
```

AVG_PAYLOAD_MASS__KG_
2534.6666666666665

First Successful Ground Landing Date

- This date represents one of the most critical inflection points in aerospace history and our dataset. On 22 December 2015, SpaceX achieved its first-ever successful return-to-launch-site ground landing.

```
%sql SELECT MIN(Date) from SPACEXTABLE WHERE "Landing_Outcome" = 'Success (ground pad)';
```

```
 * sqlite:///my_data1.db
Done.
```

MIN(Date)
2015-12-22

Successful Drone Ship Landing

- Boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000.
- The earlier F9 v1.1 cores could not land drone-ship missions with payloads above 4,000 kg. It wasn't until the introduction of the Falcon 9 Full Thrust (FT) version of the B1 cores that this became possible.

```
%%sql
SELECT "Booster_Version", PAYLOAD_MASS__KG_, "Landing_Outcome" from SPACEXTABLE
    WHERE "Landing_Outcome" = 'Success (drone ship)'
    AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000;
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version	PAYLOAD_MASS__KG_	Landing_Outcome
F9 FT B1022	4696	Success (drone ship)
F9 FT B1026	4600	Success (drone ship)
F9 FT B1021.2	5300	Success (drone ship)
F9 FT B1031.2	5200	Success (drone ship)

Total Successful and Failed Mission Outcomes

- The overwhelming ratio of successful mission outcomes directly correlates with SpaceX's market dominance and why the Falcon 9 disrupted the entire aerospace industry.

```
%sql SELECT "Mission_Outcome", COUNT(*) AS TOTAL_OUTCOMES from SPACEXTABLE GROUP BY "Mission_Outcome";
```

```
 * sqlite:///my_data1.db
Done.
```

Mission_Outcome	TOTAL_OUTCOMES
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carrying Maximum Payload

- We can identifiy the Falcon 9 Block 5 (B5) as the final, most optimized iteration of the B1 core, achieving the maximum payload capacity of **15,600 kg**.

```
%%sql
SELECT "Booster_Version", PAYLOAD_MASS__KG_ from SPACEXTABLE
    WHERE PAYLOAD_MASS__KG_ = (select MAX(PAYLOAD_MASS__KG_) from SPACEXTABLE);
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version	PAYLOAD_MASS__KG_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

2015 Launch Records

- Failed drone ship landing outcomes with the corresponding booster version and launch site for months in the year 2015.
- The January and April 2015 outcomes map out the rapid troubleshooting loops required to perfect rocket recovery.

```
%%sql
SELECT
    CASE substr(Date, 6, 2)
        WHEN '01' THEN 'January'
        WHEN '02' THEN 'February'
        WHEN '03' THEN 'March'
        WHEN '04' THEN 'April'
        WHEN '05' THEN 'May'
        WHEN '06' THEN 'June'
        WHEN '07' THEN 'July'
        WHEN '08' THEN 'August'
        WHEN '09' THEN 'September'
        WHEN '10' THEN 'October'
        WHEN '11' THEN 'November'
        WHEN '12' THEN 'December'
    END AS "Month_Name",
    "Landing_Outcome",
    "Booster_Version",
    "Launch_Site"
from SPACEXTABLE
WHERE "Landing_Outcome" = 'Failure (drone ship)'
  AND substr(Date, 0, 5) = '2015';
```

```
 * sqlite:///my_data1.db
Done.
```

Month_Name	Landing_Outcome	Booster_Version	Launch_Site
January	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
April	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Ranking Landing Outcomes

- Ranking the count of landing outcomes between 2010-06-04 and 2017-03-20, in descending order.
- 'No Attempt' dominates the early phase of this specific timeframe. This highlights that for the first several years of the Falcon 9 program, reusability was strictly theoretical. SpaceX's primary contractual obligation was simply payload delivery. The data reflects a deliberate baseline phase where structural retrieval was not factored into mission parameters.

```
%%sql
SELECT "Landing_Outcome", COUNT(*) AS Outcome_Count FROM SPACEXTABLE
    WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
    GROUP BY Landing_Outcome ORDER BY Outcome_Count DESC;
```

```
 * sqlite:///my_data1.db
Done.
```

Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1


Section 3
Launch Sites
Proximities Analysis

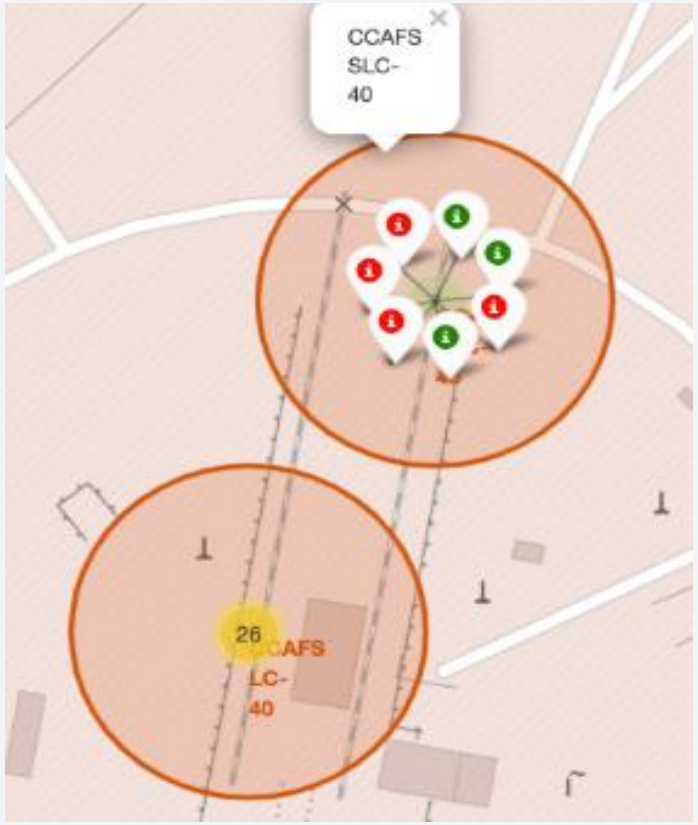
Launch Site Locations

- We can observe that all site locations are in very close proximity to the coast and are placed reasonably closer to the equator for enhanced safety and flight trajectory.



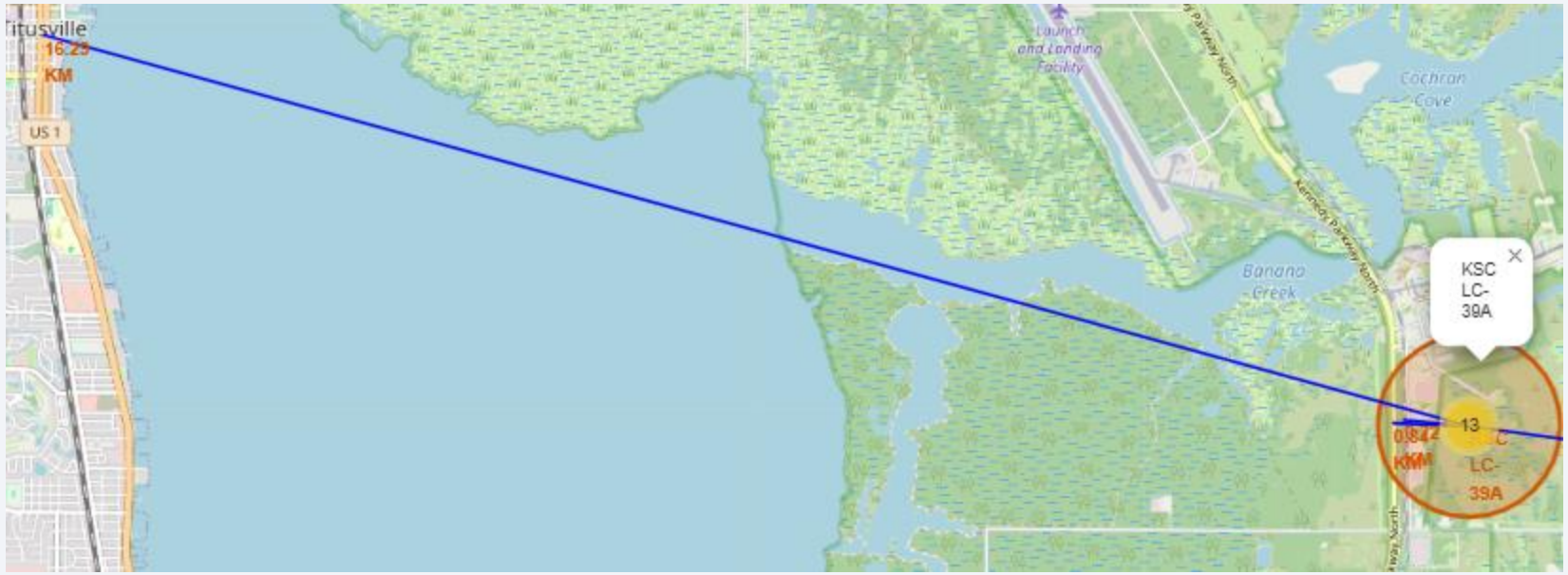
Color-labeled Launch Outcomes

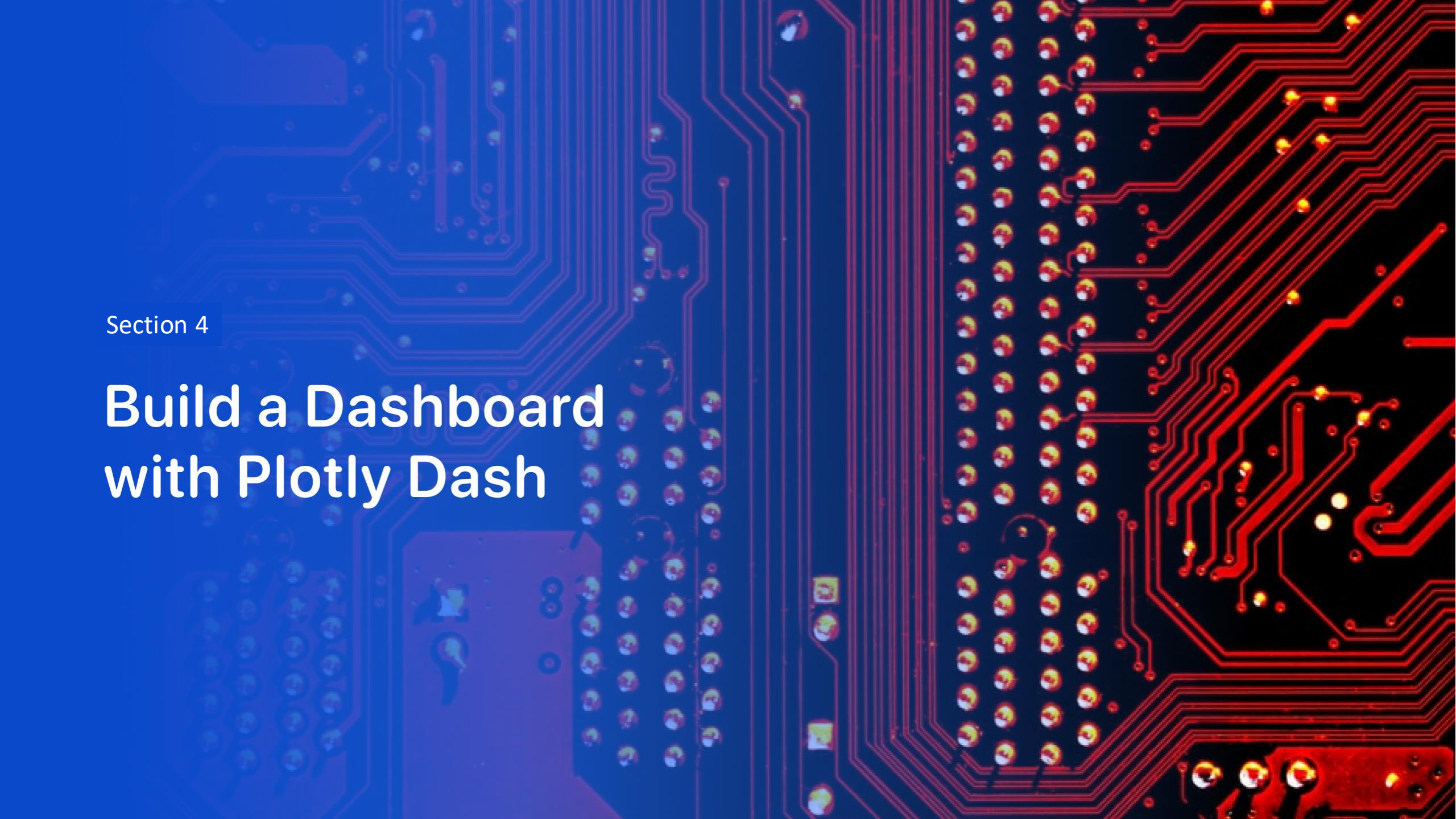
- Each site contains color-labeled launch outcomes. Red=Failed, Green=Successful.



Launch Site Proximities

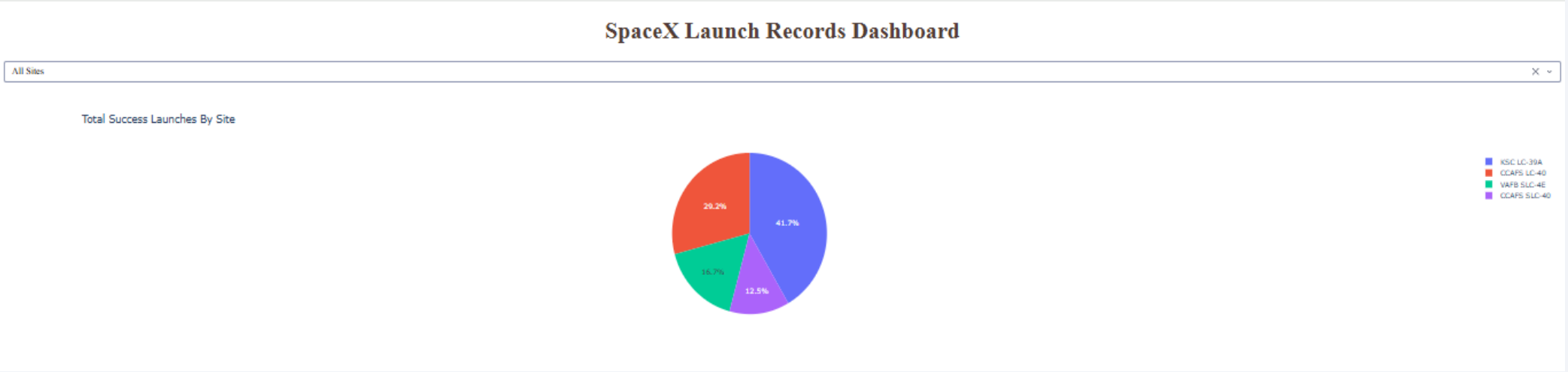
- All launch sites are in close proximity to coastlines, railways, highways and maintain a safe distance from cities.
- KSC LC-39A is less than 1km from the nearest railway and highway, less than 4km from the nearest coastline and roughly 16km away from the nearest city.




Section 4
Build a Dashboard with Plotly Dash

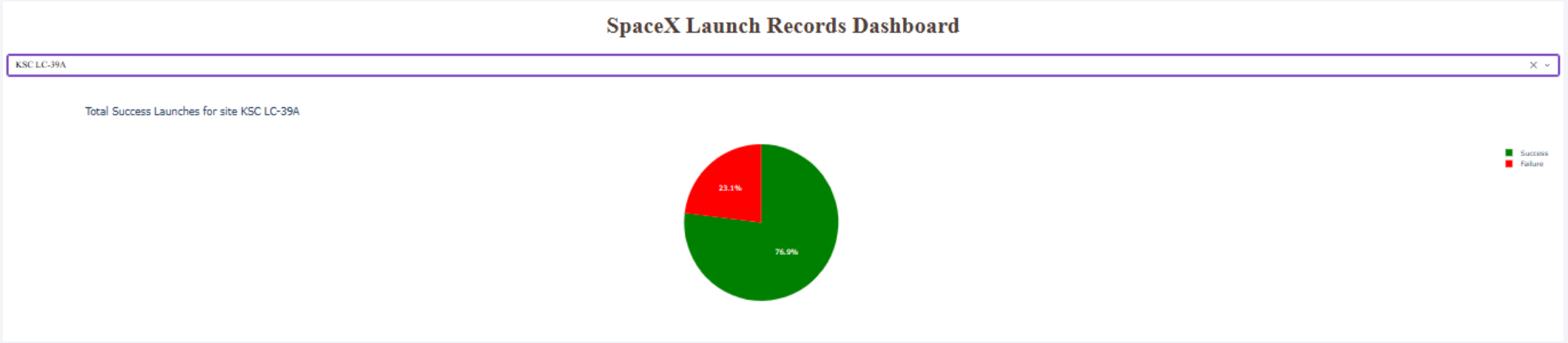
Launch Success Count

- When the dashboard dropdown is set to 'All Launch Sites', the pie chart renders total volume distribution which displays KSC LC-39A as the top success producer at **41.7%**.



Highest Launch Success Ratio

- KSC LC-39A achieves the highest individual success ratio in the dataset at **76.9%**.
- The high success ratio directly correlates with the introduction of highly reliable booster configurations, specifically the Falcon 9 Full Thrust and Block 5 iterations.



Payload Range vs Launch Outcome

- Low-to-medium payloads (**2,000 kg – 5,300 kg**) yield highly consistent success rates due to generous booster propellant margins.
- Payloads in the **5,300 kg – 7,000 kg** range mark a historical phase of volatility, where tight fuel margins required high-risk, high-velocity downrange drone ship recoveries.
- The **Falcon 9 Block 5** booster version commands a 100% success rate at maximum payloads (**13,000+ kg**), demonstrating the absolute maturity of the hardware architecture.



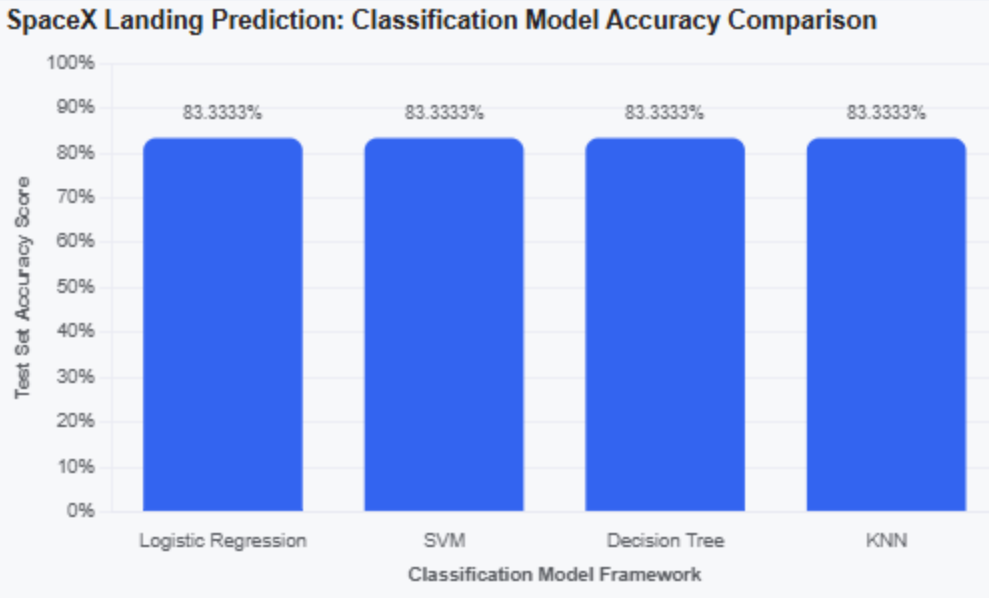
Section 5

Predictive Analysis (Classification)



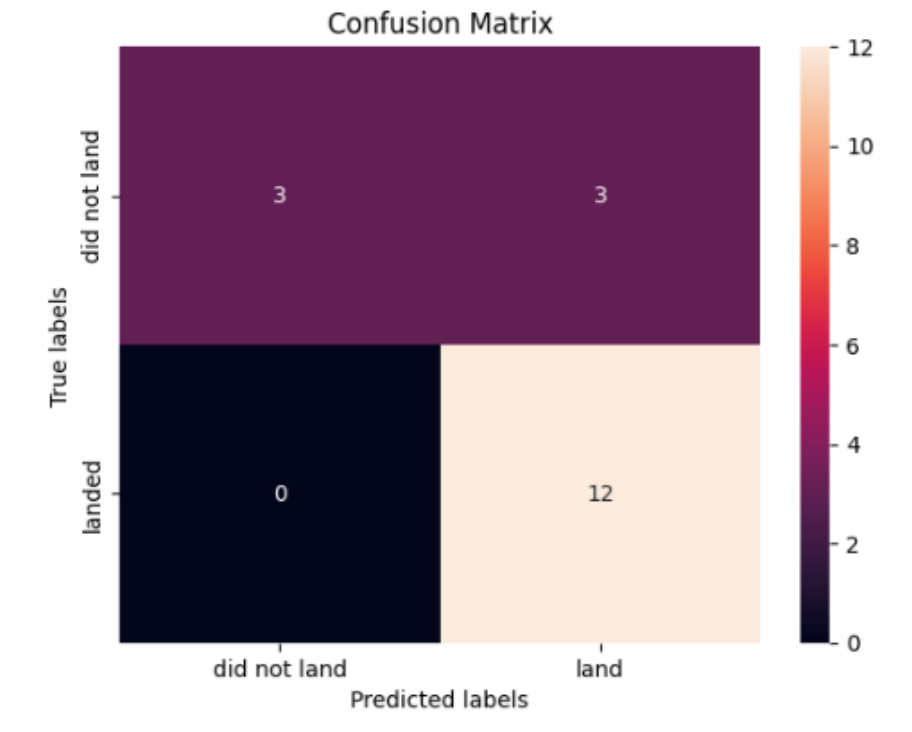
Classification Accuracy

- The uniform accuracy score of **83.33%** is structurally driven by the small test set partition.



Confusion Matrix

- All four classification models produce the same confusion matrix with a true positive of 12 and false positive of 3.



Results: EDA Visualisations

- **Flight Number vs. Success Trend:**
  - Flights 1–40 (Experimental Era) show a high density of failures.
  - Flights 50–90 (Matured Era) show a dense, continuous band of successful outcomes.

- **Orbit Profiles:** Low-altitude paths like **LEO, VLEO, ISS, and SSO** exhibit high success rates.

- **High-Risk Profiles:** Geostationary Transfer Orbits (**GTO**) present a mixed success ratio, driven by extreme thermal and atmospheric stress during high-velocity re-entry profiles.

Results: EDA with SQL

- **Launch Site Volume Performance:** SQL aggregation queries pinpointed KSC LC-39A and CCAFS LC-40 as the highest volume launch pads.
- **Historical Milestones:** Isolated booster core ID B1021 via targeted SQL queries as the first rocket to complete a successful recovery on an Autonomous Spaceport Drone Ship (ASDS).
- **Payload Segment Identification:** SQL filtering isolated a distinct operational window for medium-to-heavy commercial payloads between 4,000 kg and 6,000 kg, tracking their recovery rates across launch locations.

Results: Interactive Map with Folium

- **Coastal Optimisation:** All active pads sit within **1 km of the coastal line**, ensuring launch trajectories clear populated areas to protect civilian lives.
- **Logistics Integration:** Maps show that launch pads are placed near critical commercial rail networks, which are required to transport heavy booster stages from manufacturing centers.
- **Buffer Clearances:** Maintained a strict **15–35 km safety buffer** away from cities to isolate communities from blast radii and severe acoustic vibrations.

Results: Dashboard with Plotly Dash

- **Dynamic Site Filtering:** The dropdown interface updates the dashboard in real time, shifting from aggregate global metrics to site-specific landing success distributions.
- **Payload Capacity Clustering:**
  - Payloads below **3,000 kg** exhibit high volatility with mixed success and failure marks.
  - Payloads between **5,500 kg and 10,000 kg** form a highly reliable cluster of successful landings.
- **Booster Version Performance:** Cross-filtering highlights that older booster iterations dominate early failure zones, while the **Block 5** variant achieves excellent execution.

Results: Predictive Analysis (Classification)

- **Model Score Summary:**
  - Logistic Regression: 83.33% Test Accuracy
  - Support Vector Machine (SVM): 83.33% Test Accuracy
  - K-Nearest Neighbors (KNN): 83.33% Test Accuracy
  - Decision Tree Classifier: 83.33% Test Accuracy (87.50% Training Cross-Validation)
- **The Small-Sample Paradox:** All four classifiers tied with an 83.33% score on the 18-record test subset, producing identical confusion matrices.
- **Model Selection Decision: Logistic Regression** and **SVM** are selected as the optimal models for production deployment. They offer better generalisation and lack the high-variance overfitting risks seen in the Decision Tree.

Conclusion

- **Geospatial Layout Strategy:** Launch pads are deliberately optimized inside a narrow 1 km coastal corridor near commercial rail spurs to streamline heavy booster logistics while safely maintaining a 15–35 km buffer away from major population centers.
- **Hardware Evolution Impact:** The transition from early experimental failures to near-flawless recovery records is directly driven by the introduction of the Falcon 9 Block 5 architecture, which optimized engine configurations and introduced durable titanium grid fins.
- **Business Model Disruption:** Perfecting first-stage recoveries directly defends SpaceX's baseline $62 million pricing advantage over traditional legacy launch providers who discard their multi-million dollar rocket cores.

Thank you!

